

APP Data Structure

PHASE 1 – COLLEGE

1. Colleges Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
college_name	Text (Not Null)
address	Text (Not Null)
email	Text (Not Null)
phone	Text (Not Null)
contact_person_1_name	Text (Not Null)
contact_person_1_phone	Text (Not Null)
contact_person_2_name	Text
contact_person_2_phone	Text
contact_person_3_name	Text
contact_person_3_phone	Text
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

2. Batches Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
student_id	UUID (Foreign Key → students.id)
batch_year	Text (Not Null)
start_month	SmallInt (Not Null, 1–12)
end_month	SmallInt (Not Null, 1–12)
target_hours_min	Integer, Default 300
target_hours_max	Integer, Default 350
batch_status	Text, One of ('planning','active','completed','archived')
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

PHASE 2 – STUDENT

3. Students Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
student_name	Text (Not Null)
email	Text
phone	Text
student_id	UUID (Foreign Key → students.id)
course	Text (Not Null) — e.g. B.Com, BCA, BA, MSc
parent_guardian_name	Text (Not Null)

parent_guardian_phone	Text (Not Null)
parent_guardian_email	Text
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

4. Panel Members Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
panel_member_name	Text (Not Null)
email	Text (Not Null, Unique)
phone	Text (Not Null)
address	Text (Not Null)
photograph_url	Text (Not Null)
pan_number	Text (Not Null)
linkedin_url	Text (Not Null)
pay_per_hour	Numeric(10,2) (Not Null), Default 0
visit_allowance	Numeric(10,2), Default 0.00
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

5. Entrance Score Weights Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
batch_id	UUID (Foreign Key → batches.id)
university_marks_weight	Numeric(5,4)
attendance_weight	Numeric(5,4)
lgiver_apititude_weight	Numeric(5,4)
hod_feedback_weight	Numeric(5,4)
panel_member_1_weight	Numeric(5,4)
panel_member_2_weight	Numeric(5,4)
panel_member_3_weight	Numeric(5,4)
effective_from	Date (Not Null)
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

6. Entrance Exam Scores Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
batch_id	UUID (Foreign Key → batches.id)
student_id	UUID (Foreign Key → students.id)
year1_attendance	Numeric(5,2)
year1_average_score	Numeric(5,2)
year2_attendance	Numeric(5,2)
year2_average_score	Numeric(5,2)
lgiver_apititude_score	Numeric(5,2)
hod_feedback_score	Numeric(5,2)
panel_member_1_id	UUID (Foreign Key → panel_members.id)
panel_member_1_score	Numeric(5,2)

panel_member_1_feedback	Text
panel_member_2_id	UUID (Foreign Key → panel_members.id)
panel_member_2_score	Numeric(5,2)
panel_member_2_feedback	Text
panel_member_3_id	UUID (Foreign Key → panel_members.id)
panel_member_3_score	Numeric(5,2)
panel_member_3_feedback	Text
family_income	Numeric(12,2)
Totalweighted_score	Numeric(6,2)
Selected	Boolean (Yes/No)
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

7. Student Selection & Onboarding Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
batch_id	UUID (Foreign Key → batches.id)
student_id	UUID (Foreign Key → students.id)
entrance_rank	Integer
Track	Zoho/nonZoho
parent_income_certificate_url	Text — scanned signed PDF (< ₹4 lakh income)
deposit_amount	Numeric(10,2)
deposit_received_date	Date
deposit_receipt_url	Text — PDF receipt when deposit collected
deposit_refund_date	Date — refunded on course completion
course_status	Text, One of ('in_progress','completed','dropout')
course_status_description	Text — reason if dropout (e.g. got job, left country)
certificate_status	Text, One of ('not_issued','issued')
certificate_id	UUID (Foreign Key → certificates.id)
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()

8. Student Evaluation Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
college_id	UUID (Foreign Key → students.college_id, Not Null)
subject_id	UUID (Foreign Key → subjects.id, Not Null)
module_id	UUID (Foreign Key → modules.id, Not Null)
evaluation_name	Text (Not Null)
score	Numeric(6,2) (Not Null)
max_score	Numeric(6,2) (Not Null)
entered_by	UUID (Foreign Key → trainers.id, Not Null)
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

9. Certificates Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
certificate_name	Text (Not Null)

description	Text
template_url	Text
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()

PHASE 3 – TRAINER

10. Trainers Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
user_id	UUID (Foreign Key → users.id, Unique)
trainer_name	Text (Not Null)
email	Text (Not Null, Unique)
phone	Text (Not Null)
address	Text (Not Null)
photograph_url	Text (Not Null)
pan_number	Text (Not Null)
linkedin_url	Text (Not Null)
pay_per_hour	Numeric(10,2) (Not Null), Default 0
visit_allowance	Numeric(10,2), Default 0.00
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

11. Subject Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
trainer_id	UUID (Foreign Key → trainers.id, Not Null)
subject_name	Text (Not Null, Unique) — e.g. JavaScript, HTML, CSS, Zoho Creator
description	Text
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

12. Modules Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
subject_id	UUID (Foreign Key → subject.id, Not Null)
module_name	Text (Not Null) — e.g. JS Syntax
description	Text — scope, reading links, introductory vs in-depth notes
allocated_hours	Numeric(5,2) (Not Null)
sort_order	SmallInt, Default 0
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

13. Topic Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
module_id	UUID (Foreign Key → modules.id, Not Null)
topic_name	Text (Not Null) — e.g. JS Statements, JS Comments
description	Text — multi-line paragraph; scope, reading material, video links
duration_hours	Numeric(4,2) (Not Null, 1.00–2.00)
sort_order	SmallInt, Default 0
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

14. Trainer Batch Mapping Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
trainer_id	UUID (Foreign Key → trainers.id)
batch_id	UUID (Foreign Key → batches.id)
assigned_at	Timestamp with Time Zone, Default now()
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

PHASE 4 – TRAINING EXECUTION

15. Availability Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
batch_id	UUID (Foreign Key → batches.id)
available_date	Date (Not Null)
start_time	Time (Not Null)
end_time	Time (Not Null)
total_hours	Numeric(4,2) (Not Null, > 0)
class_type	Text, Must be 'Lab' or 'Classroom'
slot_status	Text, One of ('available', 'mapped', 'consumed')
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

16. Class Session Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
batch_id	UUID (Foreign Key → batches.id)
trainer_id	UUID (Foreign Key → trainers.id)
topic_id	UUID (Foreign Key → topic.id)
availability_id	UUID (Foreign Key → availability.id, Unique when active)
planned_class_mode	Text, One of ('Online Lab', 'Online Classroom', 'Offline Lab', 'Offline Classroom')
class_status	Text, One of ('pending', 'conducted', 'not_conducted')
actual_start_time	Time
actual_end_time	Time
actual_class_mode	Text, One of ('Online Lab', 'Online Classroom', 'Offline Lab', 'Offline Classroom')

attendance_register_taken	Boolean
trainer_remarks	Text
admin_remarks	Text
reschedule_required	Boolean
mapped_by	UUID (Foreign Key → users.id)
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

17. Assessment Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
module_id	UUID (Foreign Key → modules.id, Not Null)
trainer_id	UUID (Foreign Key → users.id, Not Null)
assessment_date	Date (Not Null)
assessment_type	Text (MCQ / Coding / Practical / Type A / Type B / Type C)
question_document	Text (Question Document / Excel File Path / URL)
assessment_notes	Text (Instructions / Additional Notes)
total_marks	Numeric(5,2), Default 100
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

18. Roles Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
role_name	Text (Not Null, Unique)
description	Text
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

19. Users Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
full_name	Text (Not Null)
email	Text (Not Null, Unique)
password_hash	Text (Not Null)
phone	Text
role_id	UUID (Foreign Key → roles.id, Not Null)
last_login_at	Timestamp with Time Zone
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

PHASE 8 – PLACEMENT

20. Companies Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
company_name	Text (Not Null, Unique)
contact_person	Text
contact_email	Text
contact_phone	Text
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

21. Student Interview Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
student_id	UUID (Foreign Key → students.id)
company_id	UUID (Foreign Key → companies.id)
interview_date	Date (Not Null)
interview_time	Time
interview_result	Text, One of ('on_hold', 'next_round', 'cleared', 'failed', 'pairing')
notes	Text
scheduled_by	UUID (Foreign Key → users.id)
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

22. Placement Tracking Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
student_id	UUID (Foreign Key → students.id)
company_position_id	UUID (Foreign Key → company_positions.id)
placement_status	Text, One of ('not_started', 'in_progress', 'placed', 'not_placed')
offer_date	Date
acknowledgement_doc_url	Text
starting_salary	Numeric(12,2)
offer_letter_url	Text
placement_notes	Text
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

PHASE 9 – PAYMENT

23. Billing Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
trainer_id	UUID (Foreign Key → trainers.id)
billing_period_start	Date (Not Null)
billing_period_end	Date (Not Null)
invoice_number	Text

subtotal_amount	Numeric(12,2), Default 0
total_amount	Numeric(12,2), Default 0
billing_status	Text, One of ('draft','pending_approval','approved','paid','rejected')
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()

PHASE 10 – NOTIFICATIONS & SYSTEM

24. Notification Templates Table

Field Name	Data Type / Constraints
id	UUID (Primary Key)
event_type	Text (Not Null)
channel	Text, One of ('email','whatsapp')
subject	Text
body_template	Text (Not Null)
active	Boolean, Default TRUE
created_at	Timestamp with Time Zone, Default now()
updated_at	Timestamp with Time Zone, Default now()